

Moment Retrieval

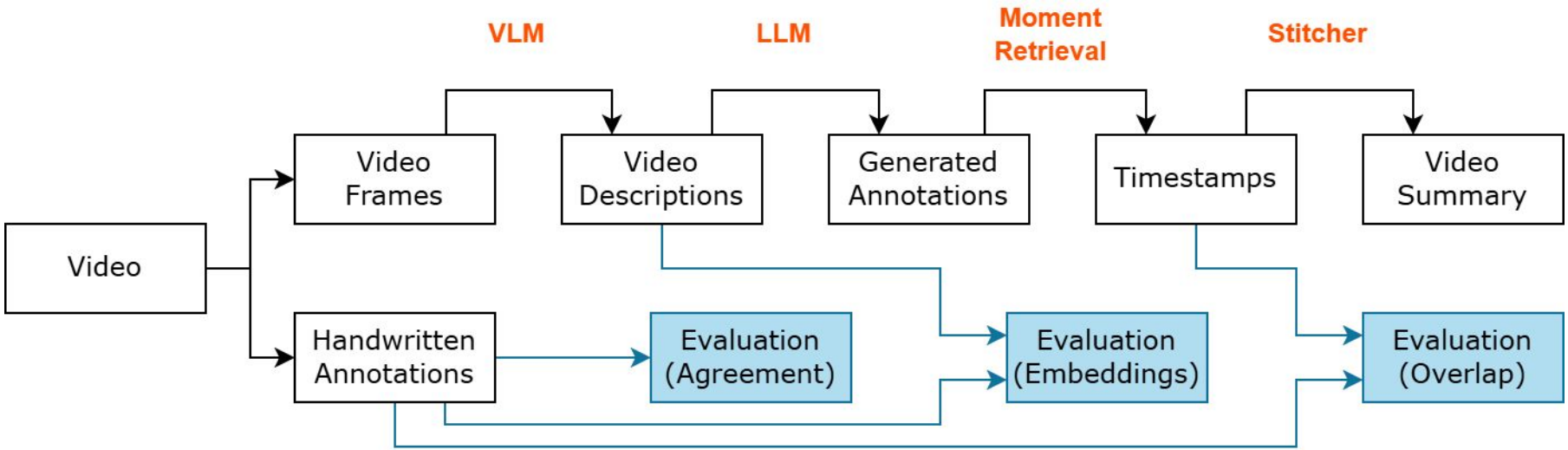
Presented by Group 7

Computer Vision

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Pipeline in Progress



Annotations

Videos

- TVSum
- 25, 26, 27, 28

Challenges

- Chaotic videos (25 requires multiple watches to understand)
- Repetition of many similar scenes
- Video length (more than 150 seconds long)
- Relying on audio heavily (especially video 27)



VLM + LLM

Generating the video descriptions

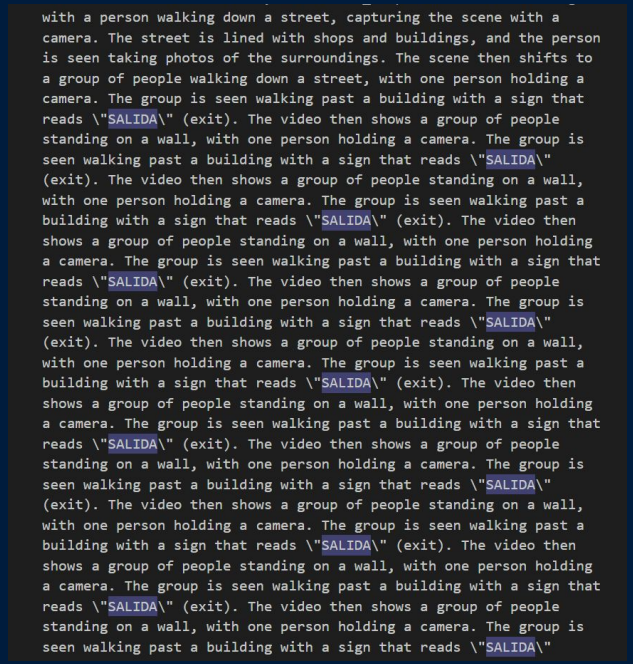
- SmolVLM 2
- Takes n frames + prompt; outputs of the events of the video

Generating the annotations

- Llama 3.2 (+ all-MiniLM-L6-v2)
- Takes VLM's description, outputs formatted, meaningful descriptions

Challenges

- Repetition loop (repetition_penalty, no_repeat_ngram_size)
- No timestamps available



Moment Retrieval: QD-DETR and CG-DETR

Takes the list of events as input, outputs the best matching timestamps

Results

- Struggles with repeated events and short moments

Challenges

- Problem: Videos less than 150 seconds
  - What helped? Speed-up the video
- Problem: Moments are often 10-20 seconds long
  - What helped? Rerun the model on the retrieved moment

Moment Retrieval: Video Feature Representations

Feature encoding

- **CLIP -** Can identify objects on individual frames
- **CLIP + SlowFast -** Is better at modeling active motion throughout the video

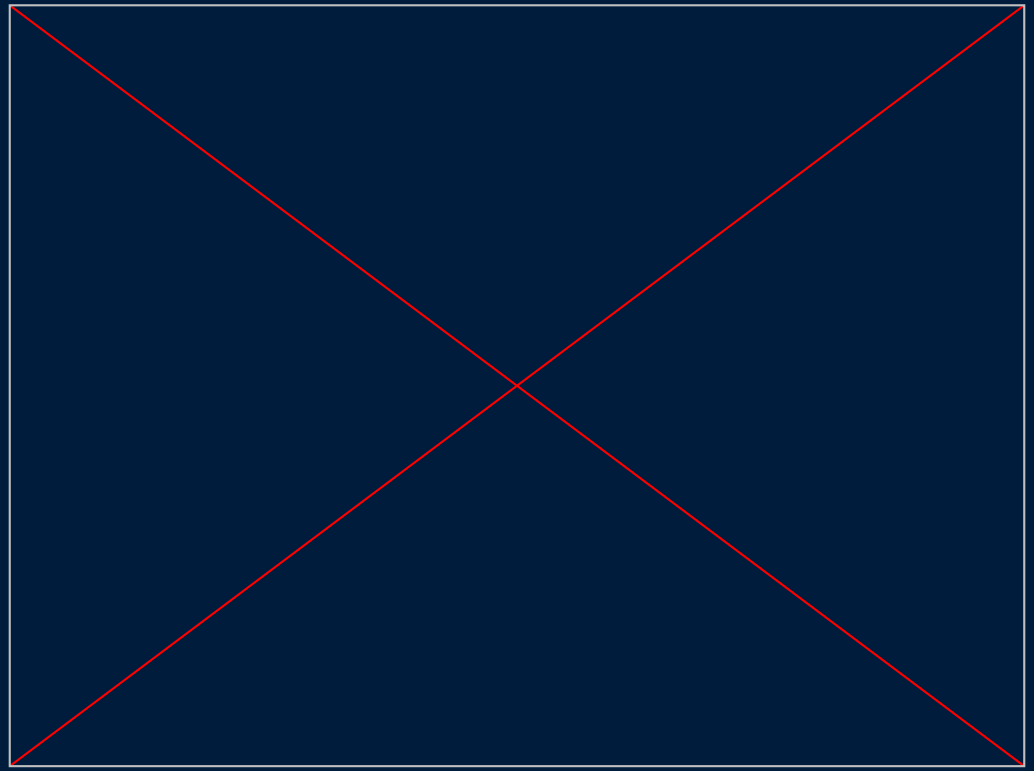
Observations

- Adding SlowFast improved long moments problem and resulted in higher confidence scores in outputs

Example

1. A person walking down a street in a city, passing by various buildings and shops. | Predicted Timestamp (75.1, 90.6)
2. A person sitting on the ground, wearing earphones and looking at something in their hands | Predicted Timestamp (91.4, 97.8)
3. A group of people gathered around a banner, with one person holding a microphone | Predicted Timestamp (22.0, 32.7)
4. The video then shows a group of people standing on a beach, with a large building in the background | Predicted Timestamp (172.6, 184.0)
5. A person lying on a bed, holding a camera | Predicted Timestamp (228.6, 241.6)
6. A group of people playing in a dirt field, with a large building in the background | Predicted Timestamp (189.3, 202.1)

Example





Thanks for listening!

Questions?

Presented by Group 7